

DSA-1 MID QB

TimeComplexity:

b) Discuss the time complexity of the following algorithm.

```

sum=0;
for(i=0; i<n; i++){
    scanf("%d", &A[i]);
    j=n-1;
    while(j>=0){
        sum=sum+A[i]+A[j];
        j--;
    }
}
    
```

Handwritten notes: $O(1)$, Δ , (n)

$y=n$

Handwritten calculations: $n \cdot n$, $(n+1)n$, $0, 1, 2, 3$, $+1$, 1

b) Discuss the time complexity of the following algorithm. [3]

```

sum=0;
for(i=1; i<=n; i++){
    for(j=1; j<=n; j++){
        sum=sum+i+j;
    }
}
printf("%d", sum);
    
```

Handwritten notes: (n) , $n+1$, $(n+1)n$, $n \cdot n$, $1, 2, 3, 4$, $+1$

b) Discuss the time complexity of the following algorithm. [2]

```

sum=0;
for(i=1; i<=n; i++){
    for(j=1; j<=i; j++){
        sum=sum+i+j;
    }
}
printf("%d", sum);
    
```

Handwritten notes: $i: 1, 3, 6$, $1, 2, 4, 7, \dots$, $n, (n+1), (n+2), \dots$, $\frac{n(n+1)}{2}$, $1 + n$, $\frac{(n+1)(n+2)}{2}$

② 3
b) Discuss the time complexity of the following algorithm.

```
sum=0;  
for(i=2; i<=n; i++){  
    for(j=2; j<=i; j++){  
        sum=sum+i+j;  
    }  
    printf("%d", sum);  
}
```

b) Discuss the time complexity of the following algorithm.

```
sum=0;  
for(i=1; i<=n; i++){  
    for(j=1; j<=i; j++){  
        sum=sum+i+j;  
    }  
}  
printf("%d", sum);
```

b) Discuss the time complexity of the following algorithm.

```
sum=0;  
for(i=1; i<=n; i++){  
    for(j=1; j<=n; j++){  
        sum=sum+i+j;  
    }  
}  
printf("%d", sum);
```

[3]

5

[3]

✓

end

InsertionSort:

2. a) How many times the condition of while loop in the **Ascending Order Insertion Sort Algorithm** will be executed for the following data? [3]

Insertion Sort Algorithm:
for j=2 to n do
 t=A[j]
 i=j-1
 while ((i>=1) AND (A[i]>t))
 A[i+1]=A[i]
 i=i-1
 end while
 A[i+1]=t
end for

Data Set-I: 40, 30, 20, 10
Data Set-II: 10, 20, 30, 40
Data Set-III: 30, 10, 20, 40

2. a) How many times the condition of while loop in the **Ascending Order Insertion Sort Algorithm** will be executed for the following data? [3]

Data Set-I: 10, 20, 30, 40

Data Set-II: 40, 30, 20, 10

Data Set-III: 30, 10, 40, 20

4-2

mergeSort:-----

1. a) How does the descending order merge sort algorithm work on the following data? [2]

y p z x r t

Here, x=last two digits of your student id+1, y=x+3, z=x+y, p=y+z, r=x+2, t=50

1. a) How do the ascending order Merge Sort algorithm and Quick Sort algorithm (upto first partition) work on the following data? [5]

y p z x r s

Here, x=last two digits of your student id+1, y=x+3, z=x+y, p=y+z, r=x+2, s=y+9

1. a) How does the ascending order Merge Sort algorithm work on the following data? [3]
- y p z x r s
- Here, x =last two digits of your student id+1, $y=x+3$, $z=x+y$, $p=y+z$, $r=x+2$, $s=y+9$

1. a) How does the **Descending Order Merge Sort** work on the following data? [3]
- y p z x r s
- Here, x =last two digits of your student id+1, $y=x+3$, $z=x+y$, $p=y+z$, $r=x+2$, $s=y+9$

1. a) How does the Ascending Order Merge Sort work on the following data? [2]
- y p z x r s
- Here, x =last two digits of your student id+2, $y=x+4$, $z=x+y$, $p=y+z$, $r=x+3$, $s=y+8$

quickShort:

1. a) How does the descending order Quick Sort algorithm work on the following data? [4]
- y p z x r
- Here, x =last two digits of your student id+1, $y=x+3$, $z=x+y$, $p=y+z$, $r=x+2$

- b) Apply the **Ascending Order Quick Sort Algorithm** for the following instance to find the first partitioning element. [2]

18 23 56 26 89 37 28 48

- b) How many element comparisons are needed for the following instance of the **Ascending Order Quick Sort Algorithm** to find the first partitioning element? [2]

18 23 56 26 89 37 28 48

ArrayMapping:

2. a) Find the memory location of $A[15][20]$ if $\text{loc}(A[0][0])=800+c$, where c =last three digits of your student id. Assume row-wise memory is allocated in the integer array $A[50][60]$, where each integer data is 4 bytes. [2]

2. a) Find the memory location of $A[15][20]$ if $\text{loc}(A[5][5])=8000+c$, where c =last four digits of your student id. Assume row-wise memory is allocated in the integer array $A[50][60]$, where each integer data is 4 bytes. [2]

2. a) Find the memory location of $A[15][20]$ if $\text{loc}(A[5][10])=8000+c$, where c =last four digits of your student id. Assume row-wise memory is allocated in the double array $A[50][60]$, where each double data is 8 bytes. [3]

2. a) Find the memory location of $A[40][70]$ if $\text{loc}(A[15][20])=8000+w$, where w =last four digits of your student id. Assume row-wise memory is allocated in the double array $A[80][100]$, where each double data is 8 bytes. [3]

- c) Find the memory location of $A[60][70]$ if $\text{loc}(A[15][20])=x+1200$, where x =last four digits of your student ID. Assume column-wise memory is allocated in the floating point type array $A[80][100]$, where each float data is 4 bytes. [2]

- c) Find the memory location of $A[70][60]$ if $\text{loc}(A[20][15])=x+1300$, where x =last four digits of your student ID. Assume row-wise memory is allocated in the floating point type array $A[80][100]$, where each float data is 4 bytes. [2]

$$\text{Loc} = (i - l_1) (u_2 - l_2 + 1) \times$$

Bound Notation

c) If $f(n)=kn^2+3n-2$, prove that $f(n)=O(n^2)$. Here, k =last digit of your student id+5. [2]

c) If $f(n)=kn^2-2$, prove that $f(n)=\Theta(n^2)$. Here, k =last digit of your student id+3. [3]

c) If $f(n)=kn^2-3$, prove that $f(n)=O(n^2)$. Here, k =last digit of your student id+2.

c) If $f(n)=kn-5$, prove that $f(n)=O(n)$. Here, k =last digit of your student id+4. [2]

b) If $f(n)=kn^2-5$, prove that $f(n)=\Theta(n^2)$. Here, k =last digit of your student id+2. [3]

b) If $f(n)=kn-4$, prove that $f(n)=O(n)$. Here, k =last digit of your student id+5. [2]

BinarySearch:

b) How does the Binary Search algorithm work for the following data? Also find the total number of element comparisons needed in this case. [2]

Input Data: x y z p r

Search Key=x

Here, x =last two digits of your student id, $y=x+3$, $z=x+y$, $p=y+z$, and $r=z+p$

b) How does the Binary Search algorithm work for the following data? Also find the total number of element comparisons needed in this case. [3]

Input Data: t r p z y x

Search Key=y

Here, x =last two digits of your student id, $y=x+3$, $z=x+y$, $p=y+z$, $r=z+p$, and $t=p+r$

b) How does the Binary Search algorithm work for the following data? Also find the total number of element comparisons needed in this case. [3]

Input Data: t r p z y x

Search Key=y

Here, x =last two digits of your student id, $y=x+3$, $z=x+y$, $p=y+z$, $r=z+p$, and $t=p+r$

b) How does the Binary Search algorithm work for the following data? Also find the total number of element comparisons needed in this case. [3]

Input Data: t r p z y x

Search Key=y

Here, x =last two digits of your student id, $y=x+3$, $z=x+y$, $p=y+z$, $r=z+p$, and $t=p+r$

3. a) How does the **Binary Search Algorithm** work on the following data? [2]

Input Data: t r p z y x

Search Key=y

Here, x =last two digits of your student ID, $y=x+3$, $z=x+y$, $p=y+z$, $r=z+p$, and $t=p+r$

3. a) How does the **Binary Search Algorithm** work on the following data? [2]

Input Data: t r p z y x

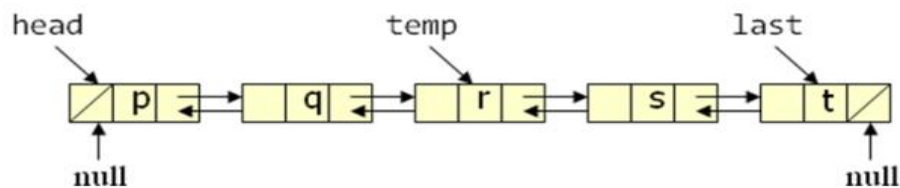
Search Key=r

Here, x =last two digits of your student ID, $y=x+4$, $z=x+y$, $p=y+z$, $r=z+p$, and $t=p+r$

linkedList:-----

3. a) Answer the following questions for the doubly linked list as shown below, where $p =$ [2.5]
last two digits of your student id + 7, $q = p+2$, $r = p+q$, $s = r-2$, $t = r+s$.

- head $\rightarrow$ next $\rightarrow$ value = ?
- last $\rightarrow$ prev $\rightarrow$ value = ?
- temp $\rightarrow$ next $\rightarrow$ next $\rightarrow$ next = ?
- temp $\rightarrow$ prev $\rightarrow$ next $\rightarrow$ value = ?
- last $\rightarrow$ prev $\rightarrow$ prev $\rightarrow$ value = ?

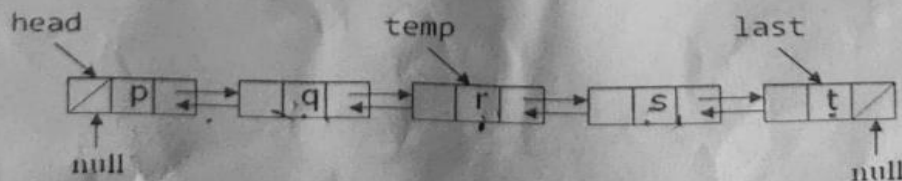


- b) Assume that you are given a single linked list as shown below. Write the statements to make it circular linked list. [1.5]



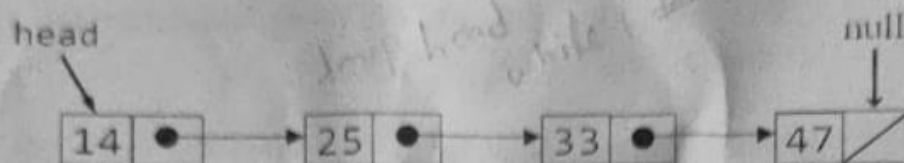
3. a) Answer the following questions for the doubly linked list as shown below, where $p =$ [3]
last two digits of your student id + 5, $q = p+3$, $r = p+q$, $s = r-1$, $t = r+s$.

- head $\rightarrow$ next $\rightarrow$ next $\rightarrow$ value = ?
- last $\rightarrow$ prev $\rightarrow$ next $\rightarrow$ value = ?
- temp $\rightarrow$ prev $\rightarrow$ prev $\rightarrow$ prev = ?
- temp $\rightarrow$ next $\rightarrow$ prev $\rightarrow$ prev $\rightarrow$ value = ?
- last $\rightarrow$ prev $\rightarrow$ prev $\rightarrow$ next $\rightarrow$ value = ?



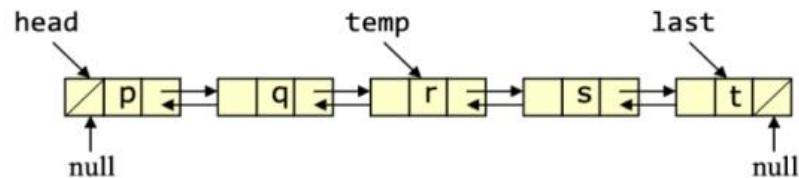
- b) Assume that you are given a single linked list as shown below. Write the statements to perform the following: [5]

- To insert 30 in between 25 and 33.
- To delete 47 from the list.
- To make a linear circular linked list from the current list.



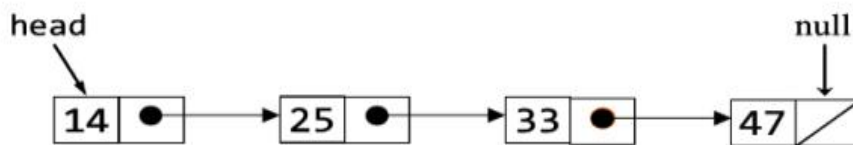
3. a) Answer the following questions for the doubly linked list as shown below, where $p =$ [3]
last two digits of your student id + 7, $q = p+4$, $r = p+q$, $s = r-3$, $t = r+s$.

- $\text{head} \rightarrow \text{next} \rightarrow \text{next} \rightarrow \text{value} = ?$
- $\text{last} \rightarrow \text{prev} \rightarrow \text{next} \rightarrow \text{value} = ?$
- $\text{temp} \rightarrow \text{prev} \rightarrow \text{prev} \rightarrow \text{prev} = ?$
- $\text{temp} \rightarrow \text{next} \rightarrow \text{prev} \rightarrow \text{prev} \rightarrow \text{value} = ?$
- $\text{last} \rightarrow \text{prev} \rightarrow \text{prev} \rightarrow \text{next} \rightarrow \text{value} = ?$



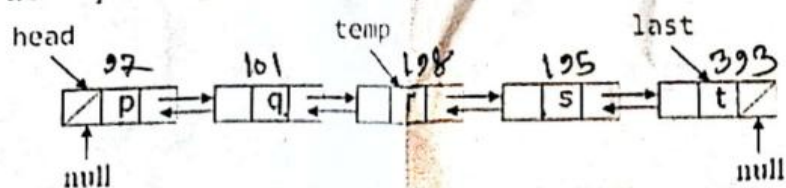
- b) Assume that you are given a single linked list as shown below. Write the statements [5]
to perform the following:

- To insert 40 in between 33 and 47.
- To delete 14 from the list.
- To make a linear circular linked list from the current list.



3. a) Answer the following questions for the doubly linked list as shown below, where $p =$ [3]
last two digits of your student id + 9, $q = p+4$, $r = p+q$, $s = r-3$, $t = r+s$.

- $\text{head} \rightarrow \text{next} \rightarrow \text{next} \rightarrow \text{value} = ?$
- $\text{last} \rightarrow \text{prev} \rightarrow \text{next} \rightarrow \text{value} = ?$
- $\text{temp} \rightarrow \text{prev} \rightarrow \text{prev} \rightarrow \text{prev} = ?$
- $\text{temp} \rightarrow \text{next} \rightarrow \text{prev} \rightarrow \text{prev} \rightarrow \text{value} = ?$
- $\text{last} \rightarrow \text{prev} \rightarrow \text{prev} \rightarrow \text{next} \rightarrow \text{value} = ?$



- b) Assume that you are given a single linked list as shown below. Write the statements [4]

to perform the following:

- To insert 40 in between 33 and 47.
- To delete 14 from the list.
- To make a linear circular linked list from the current list.



c) Suppose a linear linked list headed with "start" contains four nodes whose data values are 10, 20, 30, 40, respectively. Show the following operations. [4]

i) Draw a diagram for the linked list.

ii) Find a name for each of the nodes with respect to "start" that contain 10, 20, 30, 40, respectively?

iii) Write statements to represent 10, 20, 30, 40, respectively.

iv) Write a statement to set NULL at the end of the linked list

4. a) Show the effect of each of the statements given in the following code segment. [4]
Assume, each of the nodes in the linear linked list has two fields' **data** and **next**, where **data** is of integer type and **next** will contain the address of the next node.

```
start=(node*)malloc(sizeof(node));  
temp=(node*)malloc(sizeof(node));  
temp1=(node*)malloc(sizeof(node));  
start->data =10;  
temp->data=40;  
temp1->data=30;  
start->next=temp1;  
start->next->next=temp;  
temp->next=NULL;  
start->next=temp1->next;  
free(temp1);  
newitem=(node*)malloc(sizeof(node));  
newitem->data=34;  
newitem->next=start->next;  
start->next=newitem;
```

c) Write an algorithm to display the data stored in a double linked list in reverse order. [2]
Assume only head pointer is given for the linked list.

c) Write an algorithm to display the data stored in a double linked list in reverse order. [2]
Assume only head pointer is given for the linked list.

c) Suppose a linear linked list headed with "first" contains four nodes whose data values are 10, 20, 30, 40, respectively, where each node has two fields' **data** and **next**, where **data** is of integer type and **next** will contain the address of the next node. Show the following operations. [7]

i) Draw a diagram for the linear linked list.

ii) Find a name for each of the nodes with respect to "first" that contain 10, 20, 30, 40, respectively.

iii) Write statements to represent 10, 20, 30, 40, respectively.

iv) How can you set NULL at the end of the linked list?

v) Design a code segment to insert 35 in-between 30 and 40.

vi) How can you delete a node containing 30 from the list?

vii) Convert your linear linked list to linear circular linked list by a code segment.

4. a) Show the effect of each of the statements given in the following code segment. Assume, each of the nodes in the doubly linked list has three fields' **data**, **next** and **prev**, where **data** is of integer type, **next** and **prev** will contain the address of the next and previous nodes, respectively. [3]

```

1 start=(node*)malloc(sizeof(node));
2 temp=(node*)malloc(sizeof(node));
3 temp1=(node*)malloc(sizeof(node));
4 start->data=40;
5 temp->data=50;
6 temp1->data=20;
7 start->next=temp1;
8 temp1->prev=start;
9 start->next->next=temp;
10 temp->prev=temp1;
11 temp1->next->prev=temp1->prev;
12 temp1->prev->next=temp1->next;
13 free(temp1);
14 start->prev=NULL;
15 temp->next=NULL;

```

Stack:-----

4. a) Show the status of a STACK implemented by an array of size, $m=2$ for the operations given below. Here, $x=\text{last digit of your student id}+1$, $y=x+1$, and $z=y+2$. [2]

Push($x+y$), Push($y+z$), Pop(), Push($y*z$), Push($x*y$), Pop(), Pop(), Pop()

c) Evaluate the postfix expression, $a\ b\ +\ c\ d\ /\ -$ for $a=\text{last digit of your student id}+3$, $b=a+1$, $c=a+b$ and $d=1$ using STACK. [2]

4. a) Show the status of a STACK implemented by a linear linked list for the operations [2]
given below. Here, $x = \text{last digit of your student id} + 5$, $y = x + 3$, and $z = y + x$.

~~Pop()~~, Push($x+y$), Push($y+z$), Pop(), Push($y*z$), Push($x*y$), Pop(), Pop()

- c) Show the manual tracing of the following algorithm using a Stack;

```
#include<stdio.h>
#include<string.h>
int top=-1;
char Stack[6]={'\0'};
int main(){
    char Str1[6]={'\0'};
    char Str2[6]={'\0'};
    int i;
    strcpy(Str1, "WORLD");
    for(i=0; i<5; ++i){
        Push(Str1[i]);
    }
    for(i=0; i<5; ++i){
        Str2[i]=Pop();
    }
    printf("%s", Str2);
    return 0;
}

void Push(char x){
    Stack[++top]=x;
    return;
}

char Pop(void){
    return Stack[top--];
}
```


4. a) Show the status of a STACK implemented by a linear linked list for the operations given below. Here, x =last digit of your student id+5, $y=x+3$, and $z=y+x$. [2]

Push($x+y$), Push($y+z$), Pop(), Push($y*z$), Push($x*y$), Pop(), Pop()

- b) Show the effect of each the statements given in the following code using a Stack. [3]

```
#include<stdio.h>
#include<string.h>
int top=-1;
char Stack[4]={'\0'};
int main() {
    char Str1[4]={'\0'};
    char Str2[4]={'\0'};
    int i;
    strcpy(Str1, "CSE");
    for(i=0; i<3; ++i){
        Push(Str1[i]);
    }
    for(i=0; i<3; ++i){
        Str2[i]=Pop();
    }
    printf("%s", Str2);
    return 0;
}

void Push(char x){
    Stack[++top]=x;
    return;
}

char Pop(void){
    return Stack[top--];
}
```

4. a) Show the status of a STACK implemented by a linear linked list for the operations given below. Here, x =last digit of your student id+5, $y=x+3$, and $z=y+x$. [2]

Push($x+y$), Push($y+z$), Pop(), Push($y*z$), Push($x*y$), Pop(), Pop()

- b) Show the status of a STACK implemented by a linear linked list for the operations given below. Here, x =last digit of your student id+5, $y=x+3$, and $z=y+x$. [2]

Push($x+y$), Push($y+z$), Pop(), Push($y*z$), Push($x*y$), Pop(), Pop()

- b) Show the status of a STACK implemented by a linear linked list for the operations given below. Here, x =last digit of your student id+4, $y=x+5$, and $z=y+x$. [2]

Push($x+y$), Push($y+z$), Pop(), Push($y*z$), Push($x*y$), Pop(), Pop(), Push($x+z$)

1 2 3 4 5 6 7 8

Queue:-----

b) Show the status of a Queue implemented by a linear (singly) linked list for the following operations. Here, Enqueue and Dequeue mean insert and delete respectively, and $x = \text{last digit of your student id} + 2$, $y = x + 1$, and $z = y + 2$. [2]

Enqueue($x+y$), Enqueue($y+z$), Dequeue(), Dequeue(), Dequeue()

✓) Show the status of a Queue implemented by an array of size, $m=2$ for the following operations. Here, Enqueue and Dequeue mean insert and delete respectively, and $x = \text{last digit of your student id} + 4$, $y = x + 1$, and $z = y + x$. [2]

Enqueue($x+y$), Enqueue($y+z$), Dequeue(), Enqueue($x+y$), Dequeue(), Dequeue()

d) Show the status of a QUEUE of size 3 implemented by an array for the operations given below. Here, $x = \text{last digit of your student id} + 5$, $y = x + 3$, and $z = y + x$. Here, Enqueue and Dequeue are meant by insertion and deletion, respectively. [2]

Enqueue($x+y$), Enqueue($y+z$), Dequeue(), Enqueue($y*z$), Enqueue($x*y$), Dequeue()

c) Show the status of a QUEUE of size 3 implemented by an array for the operations given below. Here, $x = \text{last digit of your student id} + 5$, $y = x + 3$, and $z = y + x$. Here, Enqueue and Dequeue are meant by insertion and deletion, respectively. [2]

Enqueue($x+y$), Enqueue($y+z$), Dequeue(), Enqueue($y*z$), Enqueue($x*y$), Dequeue()

2.7.1

c) Show the status of a QUEUE of size 3 implemented by an array for the operations given below. Here, $x = \text{last digit of your student id} + 4$, $y = x + 5$, and $z = y + x$. Here, Enqueue and Dequeue are meant by insertion and deletion, respectively. [2]

Enqueue($x+y$), Dequeue(), Enqueue($y*z$), Enqueue($x*y$), Dequeue(), Enqueue($y+z$)